

ARSHIA AKHAVAN

Master's Student in Computer Science

@letmemakenewone@gmail.com @aakhavan3824@sdsu.com
arshiaakhavan.me arshia-akhavan ArshiAAkhavan

619-609-3341 San Diego, CA
0009-0007-9631-6964



EXPERIENCE

Software Engineer

Sotoon

Jan 2023 – Dec 2024

Tehran, Iran

Sotoon is a B2B cloud provider that focuses on reliable and scalable computing solutions. I worked in the K8s-as-a-Service team, designing and maintaining Sotoon's Kubernetes engine and several integrations to Sotoon's services, such as CSI, CNI, and Auto-scaler.

Key Achievements:

- Automate management of **90+** Kubernetes clusters, optimizing operational efficiency and reducing tickets by **30%**.
- Achieve a **40%** reduction in initial cluster **costs** by eliminating duplicated system components and introducing **multi-tenancy**.
- Redesign cluster network topology to enhance **security** and abstract control plane management from customers.
- Plan and Lead **Migration** of **over 100 customers'** k8s-clusters to the new infrastructure with **zero downtime**.
- Implement A/B testing framework for cluster updates, preventing **5** otherwise catastrophic failures.
- Investigate and debug a critical **data race** issue in the CSI plug-in that could have resulted in the deletion of over **70TB** of client storage data, ensuring data integrity and service reliability.
- Automate CI/CD pipeline for OS image creation, reducing new k8s version release time **from 1 week to 1 day**.

Data Platform Engineer

Digikala

Aug 2021 – Jan 2023

Tehran, Iran

Digikala is the largest e-commerce company in Iran, serving over 40 million users. As part of the Big Data & AI team, I was responsible for building and maintaining the infrastructure required to efficiently store, process, and manage the vast amounts of data generated by user activity.

Key Achievements:

- Developed a **Parallel Data Query Engine** on top of **Apache Spark**, revising data retrieval models and queries to reduce **resource usage** by 23% and **computation time** by 43%.
- Reduced API latency** by identifying blocking operations in an asynchronous workflow, optimizing the critical path to **triple requests per second (RPS)**.
- Designed and maintained a data platform using **K8s** for orchestration, **Ceph** for storage, **Airflow** for scheduling, **Spark** for batch/stream processing, and **Kafka** with **Debezium** for database synchronization.
- Implemented **GitOps**-based infrastructure management with **ArgoCD**, automating API deployments via **Kustomize** and **Helm**, reducing release cycles from 6 months to 6 hours.
- Deployed and maintained a **staging environment** and built a framework for the Data Science team, reducing their development time significantly.

DevOps Engineer

Tapsell

Nov 2019 – Aug 2021

Tehran, Iran

Tapsell is a leading AdTech company in Iran. As part of the DevOps team, I provided cloud infrastructure and platform services that enabled other technical teams to build, deploy, and scale their applications efficiently.

Key Achievements:

- Optimizing Cassandra's **Disk usage** for different workloads, resulting in 90% reduction in disk usage for some of our clusters.
- Designed and implemented a fully automated backup, validation, and restore system for our cloud-native infrastructure, replacing the legacy solution while preserving **backward compatibility** to enable seamless, incremental migration of 30TB+ of data with zero downtime.
- Built and maintained infrastructure by deploying a staging cluster on **K8s**, developing CI/CD pipelines with **GitLab CI**, automating provisioning with **Terraform** on **VMWare**, and administering Linux-based VMs to resolve network bottlenecks and improve performance.

Journal Articles

- ● ● ● ●



● ● ● ● ●